TrES-3b

R-1065

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_160521` · created 2026-07-30T12:48:13+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457529.7471 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.033
Transit depth (Rp/Rs)^2	3.75 +/- 1.26 [%]
Orbital Inclination (inc)	81.92 +/- 1.34
Airmass coefficient 1 (a1)	1.216 +/- 0.029
Airmass coefficient 2 (a2)	-0.155 +/- 0.005
Scatter in the residuals of the lightcurve fit is	2.26 %
Best Comparison Star	None
Optimal Aperture	2.82
Optimal Annulus	7.65
Transit Duration (day)	0.057 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.033 vs 0.16309 (deviation 0.93σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.02σ)
Mid-time O–C	-2.1 min
Depth SNR	5.8
Residual scatter	2.26 %

Light curve

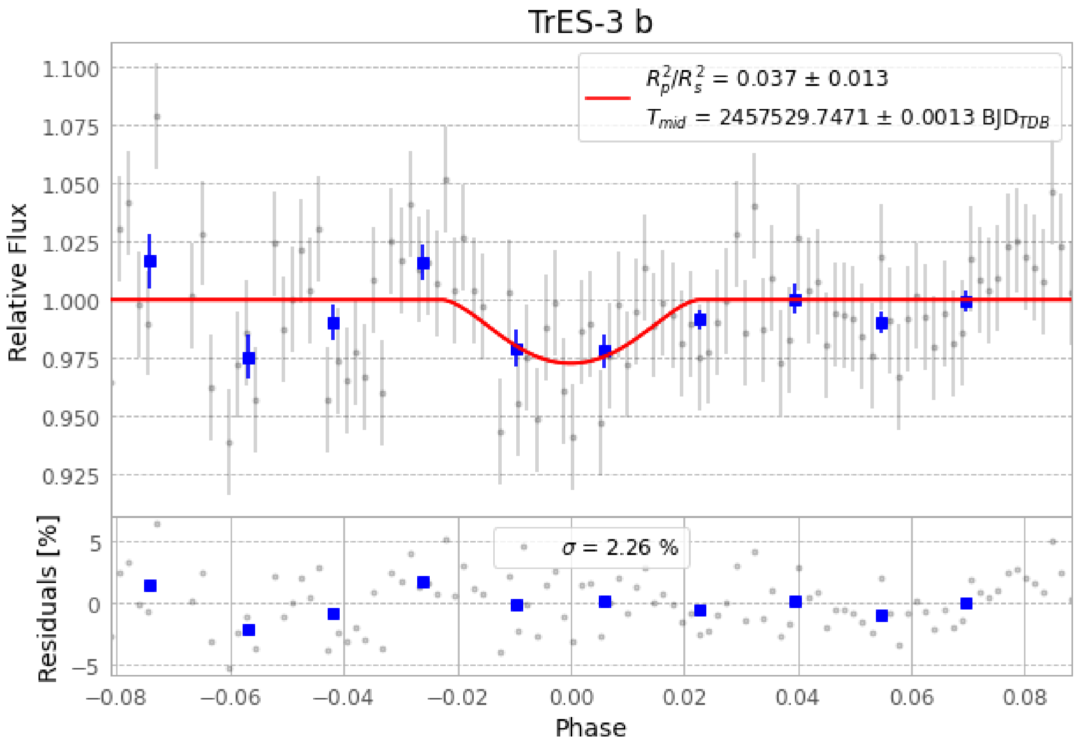

Figure 1 — FinalLightCurve_TrES-3 b_2016-05-20.png (SHA-256 26f47d2dfa789e75…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [188, 254], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 54371637368cc64d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

107 FITS frames (70 MB), `TRES-3160521031812.FITS` → `TRES-3160521083612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160521.log (12f3fffa8a2b…), FinalLightCurve_TrES-3 b_2016-05-20.png (26f47d2dfa78…), FinalLightCurve_TrES-3 b_2016-05-20.csv (af145ff255a6…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 12:48Z.